CRECENDO

Hackathon by Metapoise V2.0

Department of Computer Science and Engineering, DUIET, Dibrugarh University

Hybrid Format: Phase 1 (Online Asynchronous) ⟶ Phase 2 (On-Site Live Finals)

1. Official Competition Rules & Structure

Phase 1: Online Build & Submission Phase (Asynchronous)

Format: Remote, asynchronous prototype development and dossier submission.
Registration Fee: Free for all participating teams.

- **Problem Specificity:** Teams must strictly select and build a working prototype addressing one of the official problem statements across the **Software Track** (Section 2) or **Hardware Track** (Section 3). Unaligned open-innovation entries will not be accepted.
- **Development Window:** Teams build asynchronously from their own locations and then submit their deployed prototype link, project description, and public GitHub repository link.
- **Digital Dossier Submission:** Each team must submit a comprehensive digital package containing:
  (1) Project Description & Technical Architecture
  (2) Technology Stack & Hardware/API details
  (3) Public GitHub Repository link with clean documentation and commit history
  (4) Working Demo Link (Live Deployed Web/App URL or Hardware Bench Demonstration Video)
- **Advancement & Phase 2 Fee:** Only the top **15 selected teams** from Phase 1 will qualify for the on-site finals. Shortlisted teams will pay a registration fee of **INR 200 (Rupees 200)** to confirm their Phase 2 spot.

Phase 2: On-Site Offline Presentation & Demonstration

Eligibility: Strictly reserved for the 15 teams shortlisted from Phase 1.
Registration Fee: INR 200 per team (Applicable only to selected finalists).

- **Venue & Time:** Department of Computer Science and Engineering, DUIET, Dibrugarh University (09:00 AM – 05:00 PM).
- **Setup & Evaluation Schedule:** Teams will be provided a **2-hour setup window** (09:00 AM – 11:00 AM) to prepare their presentations and live demonstrations, after which the official evaluation by judges will commence.

Strict Disqualification Policy

As teams receive full pre-event time and technical mentorship, **if a project fails to execute or demonstrate working functionality live on stage before the jury, the team will be directly disqualified.** No simulated video rollouts or mock slides will substitute for real-time functional execution.

2. Track A: Software Problem Statements

Teams must choose **exactly one** of the official problem statements for prototype development.

PS-S01: Intelligent Flood Relief & Resource Routing System

Domain: Disaster Management | Civic Tech

The Hook: *Assam faces devastating floods annually, often leaving relief trucks and boats guessing the safest routes. Be the team that builds a digital lifeline for your state.*

Description: Develop a crowd-sourced mapping software where stranded individuals and volunteers can drop pins indicating water levels, blocked roads, and safe zones. The system should use this real-time data to generate optimized, safe routes for emergency responders and relief material distribution.

Key Feature: An offline-first or SMS-based fallback mechanism for reporting when internet connectivity drops.

PS-S02: AI-Powered Smart Contract Auditor

Domain: Blockchain | AI Security

The Hook: *Millions of dollars are lost yearly to smart contract vulnerabilities. Be the team that builds an AI guardian for Web3.*

Description: Develop an autonomous AI agent capable of analyzing Solidity smart contract code for known vulnerabilities (like reentrancy attacks or integer overflows) before deployment. The system should generate an audit report and store the audit trail immutably on a testnet blockchain.

Key Feature: Integration with a code-repository webhook for automated scanning on every pull request.

PS-S03: Predictive Maintenance IoT Edge Agent

Domain: Edge AI | Cloud

The Hook: *Factory downtime costs thousands per minute. What if machines could schedule their own repairs before breaking?*

Description: Build an edge-based AI system that ingests simulated real-time telemetry data (vibration, temperature) from industrial machines. The model should predict potential mechanical failures and automatically generate maintenance tickets to a cloud dashboard.

Key Feature: The inference must happen locally (Edge) to reduce latency, while only the anomaly alerts are synced to the cloud.

PS-S04: Smart Pothole Mapping via Passive Crowdsourcing

Domain: Smart City | Civic Tech

The Hook: *Everyone complains about bad roads and potholes, but nobody wants to fill out a long government form to report them. Automate the complaining!*

Description: Develop a mobile application that runs in the background while a user is driving or cycling. By utilizing the smartphone's accelerometer and GPS, the app should automatically detect significant bumps (potholes) and log their coordinates.

Key Feature: A public heat-map dashboard that aggregates this data to show authorities exactly which roads need the most urgent repairs.

3. Track B: Hardware Problem Statements

Teams selecting the hardware track must engineer physical embedded prototypes and bench demonstrations for **one** of the following challenges:

PS-H01: Self-Powered Agricultural Sentinel with Soil NPK Sensing

Domain: Precision AgriTech | Low-Power IoT | Embedded Systems

The Hook: *Blanket over-fertilization costs farmers billions in wasted inputs and poisons local aquifers, while soil lab testing takes weeks. Build an autonomous agricultural sentinel that monitors deep soil macronutrients and operates maintenance-free on harvested solar power.*

Description: Build a solar-powered ESP32 node that wakes up every 10 minutes, reads soil moisture, temperature, and NPK values, sends the data to a dashboard via Wi-Fi or LoRa, and returns into deep sleep. Built using off-the-shelf components: ESP32 dev board, capacitive soil moisture sensor, DS18B20 waterproof temperature probe, RS485 soil NPK sensor (with MAX485 TTL module) or analog TDS/conductivity sensor, 5V mini solar panel, TP4056 charging module, and an 18650 Li-ion battery.

Key Feature: Live dashboard updates showing telemetry when the sensor is dipped into fertilized water vs. dry soil, with hardware multimeter verification demonstrating the ESP32 successfully dropping into low-power "Deep Sleep" mode between readings.

PS-H02: Smart Wearable Fall & Cardiac Anomaly Alert Node

Domain: MedTech | Wearable Bio-Sensors | Kinematics

The Hook: *Fatal cardiac arrhythmias and traumatic falls are leading causes of sudden preventable mortality when victims lose consciousness before summoning help. Engineer a reliable wearable alert node to safeguard at-risk individuals.*

Description: Design a wearable wrist-band or chest-strap node integrating optical Pulse Oximetry (PPG) and 6-Axis Motion Kinematics. The system monitors heart rate, SpO_2, and acceleration dynamics using an ESP32 (or Seeed Studio XIAO ESP32C3), MPU6050 6-axis accelerometer/gyroscope, and MAX30102 pulse oximeter (or AD8232 ECG module), powered by a 3.7V LiPo battery.

Key Feature: Real-time autonomous alert actuation (vibrating haptic motor, active buzzer, and alert LED) and distress signal dispatch triggered upon detecting acute kinematic fall impacts or abnormal heart-rate thresholds.